

Experiment Report: Loading Effects and Impedance Matching using BJT Emitter Follower

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1 Introduction

This report investigates the phenomenon of loading effects in electronic circuits, specifically focusing on high-impedance voltage dividers. We examine how the input resistance of measurement instruments (Digital Multimeter and Oscilloscope) affects the measured output and how an Emitter Follower (Common Collector) BJT amplifier can act as a buffer to mitigate these effects.

2 Apparatus

- 5V DC Supply
- Resistors: $2 \times 1 \text{ M}\Omega$, $1 \times 100 \text{ k}\Omega \times 67 \text{ k}\Omega$
- Transistors: NPN (2N3904), PNP (2N3906)
- Breadboard and Jumper Wires
- Digital Multimeter (DMM)

3 Task 1: High-Impedance Voltage Divider

3.1 Circuit on Bread Board

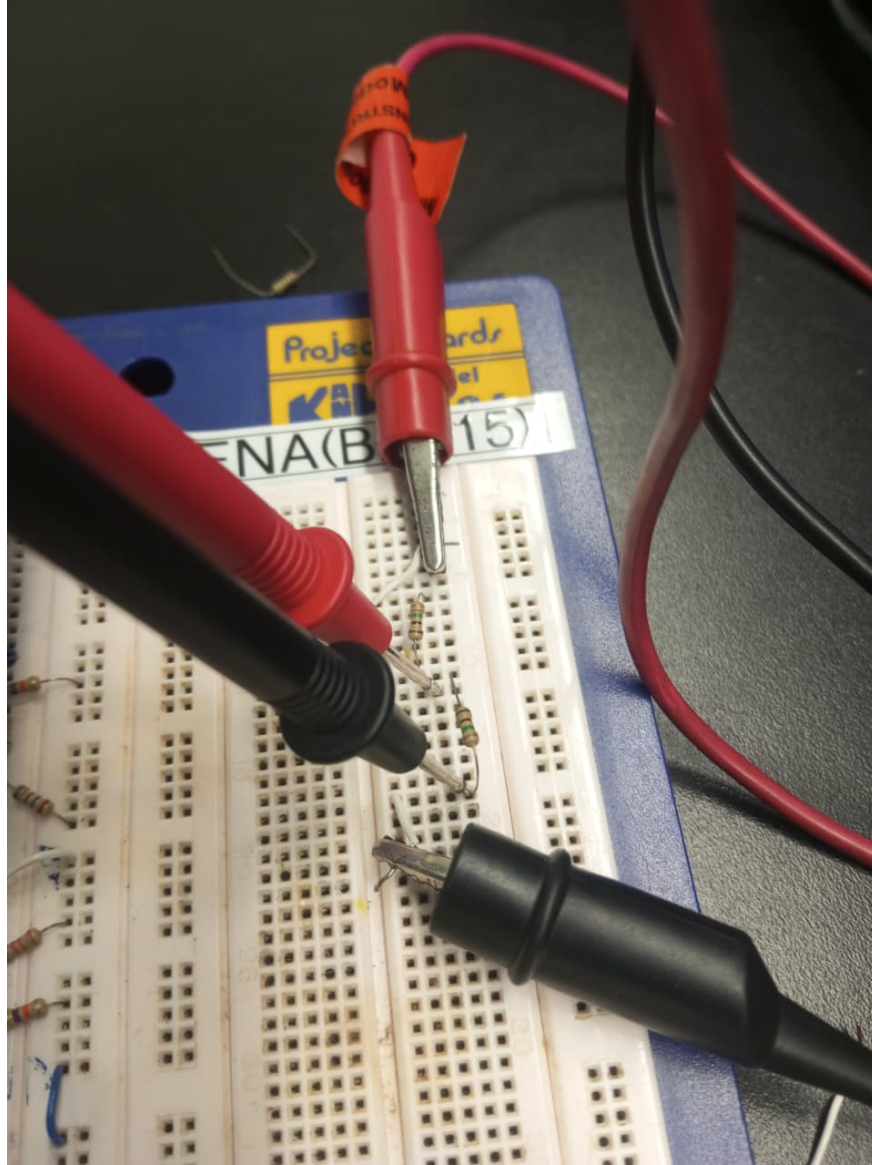


Figure 1: Voltage Divider Circuit with $R_1 = R_2 = 1\text{ M}\Omega$.

3.2 Theoretical Calculations

For an ideal divider without any external load:

$$V_{out(ideal)} = V_{in} \left(\frac{R_2}{R_1 + R_2} \right) = 5V \left(\frac{1\text{ M}\Omega}{1\text{ M}\Omega + 1\text{ M}\Omega} \right) = 2.5\text{ V} \quad (1)$$

Considering a DMM with $R_{in} \approx 10\text{ M}\Omega$:

$$R_{parallel} = \frac{R_2 \cdot R_{in}}{R_2 + R_{in}} \approx 909\text{ k}\Omega \quad (2)$$

$$V_{measured} = 5V \left(\frac{909 \text{ k}\Omega}{1000 \text{ k}\Omega + 909 \text{ k}\Omega} \right) \approx 2.38 \text{ V} \quad (3)$$

3.3 Methodology and Observation

The circuit was implemented on a breadboard. Upon measurement, the loading effect was clearly visible as the voltage dropped below the theoretical 2.5V.

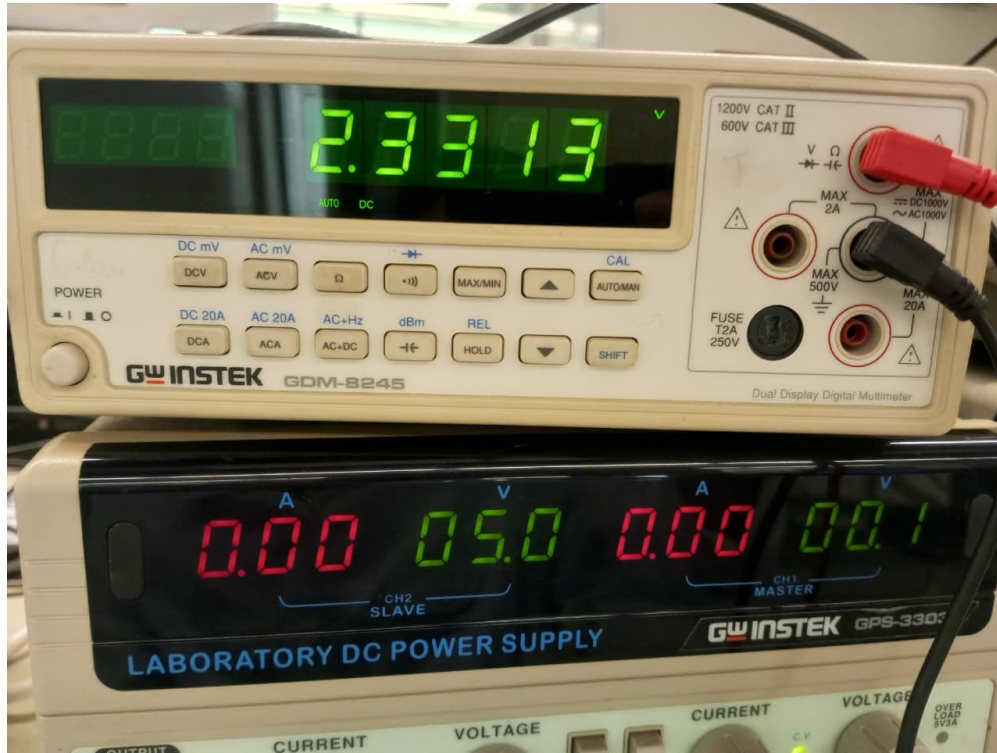


Figure 2: DMM measurement for Task 1.

4 Task 2: Emitter Follower Buffer Implementation

4.1 Circuit Diagram

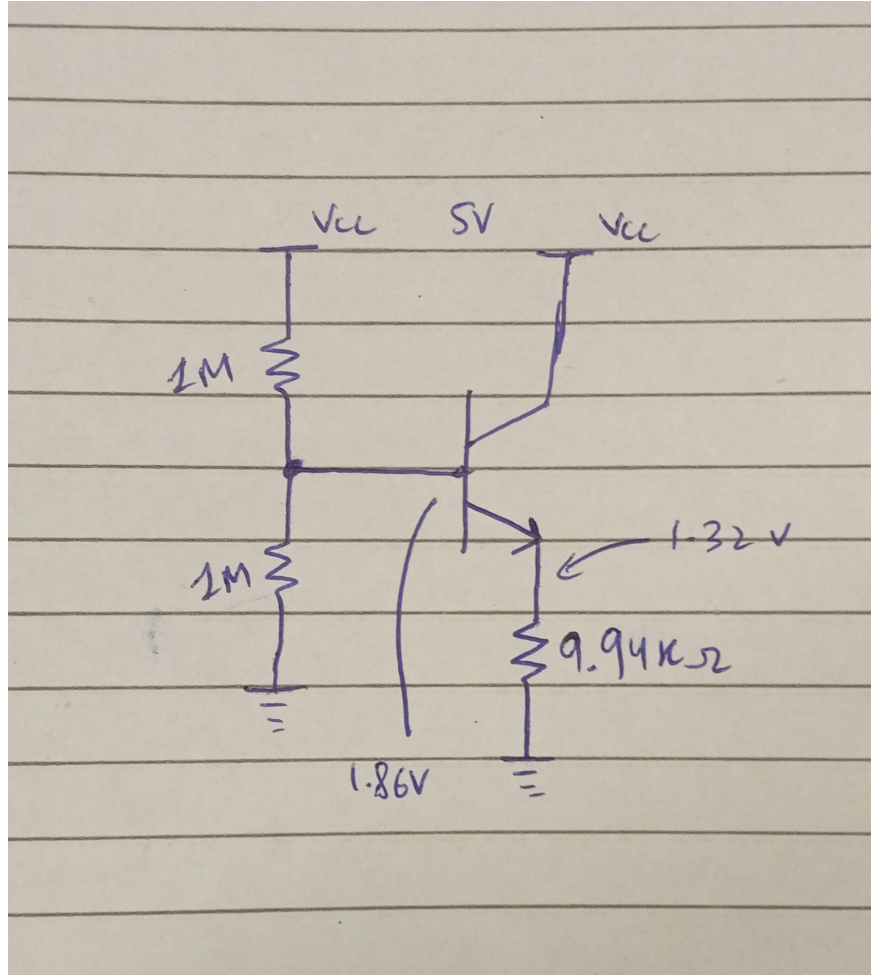


Figure 3: NPN Emitter Follower (Common Collector) Buffer Circuit.

4.2 Theoretical Analysis

The BJT provides high input impedance, isolating the divider. The output at the emitter is:

$$V_{out} = V_{base} - V_{BE} \approx 2.5V - 0.7V = 1.8V \quad (4)$$

4.3 Observations

The implementation confirms that while the voltage level is shifted by V_{BE} , the high-impedance source is no longer affected by the measurement probe's impedance. The output was observed around $\approx 1.3V$.

5 Task 3: Complementary PNP-NPN Buffer Design

5.1 Circuit Diagram

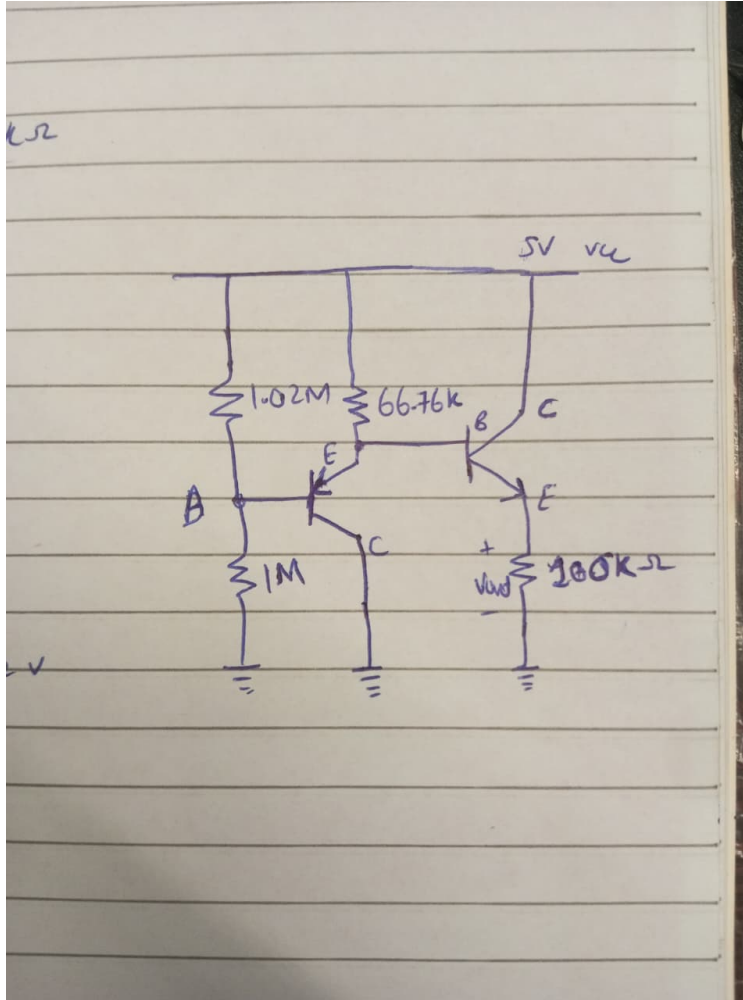


Figure 6: Hand-drawn schematic of the Complementary PNP-NPN Buffer.

5.2 Nodal Analysis and Theoretical Calculations

The design uses a two-stage complementary approach to achieve zero DC offset. The first stage (PNP) shifts the DC level up, and the second stage (NPN) shifts it back down.

Node A: Voltage Divider Output

Using the high-impedance divider ($R_1 = 1.02\text{ M}\Omega$, $R_2 = 1\text{ M}\Omega$):

$$V_A = V_{CC} \left(\frac{R_2}{R_1 + R_2} \right) = 5V \left(\frac{1\text{ M}\Omega}{1.02\text{ M}\Omega + 1\text{ M}\Omega} \right) \approx 2.4752V \quad (5)$$

Node B: PNP Emitter (First Stage)

The PNP transistor is in a common-collector configuration. Its emitter sits one V_{BE} drop above its base:

$$V_B = V_A + V_{BE(PNP)} \approx 2.4752V + 0.7171V = 3.1923V \quad (6)$$

Node Out: NPN Emitter (Second Stage)

The PNP emitter (Node B) directly drives the base of the NPN transistor. The final output is one V_{BE} drop below this base:

$$V_{out} = V_B - V_{BE(NPN)} \approx 3.1923V - 0.7V = 2.4923V \quad (7)$$

Summary of Results:

Parameter	Theoretical (Ideal)	Measured/Calculated
V_A (Node A)	2.4752 V	2.4752 V
$V_{E,PNP}$ (Node B)	3.1752 V	3.1923 V
V_{out} (Final)	2.4752 V	2.4836 V

5.3 Observations

Measured Output: 2.4836 V

Reasoning: The topology provides a nearly 1:1 voltage match to the divider ($A_v \approx 1$). The first stage provides extremely high input impedance, effectively eliminating the loading effect seen in Task 1. The small difference between the theoretical V_A (2.475V) and measured V_{out} (2.483V) is due to slight variations in the V_{BE} matching between the 2N3906 and 2N3904 transistors.

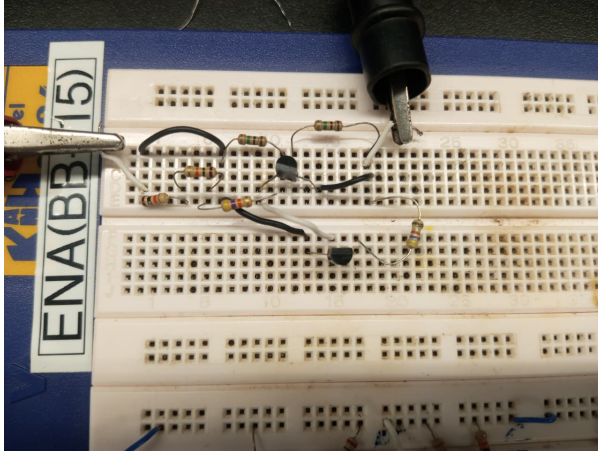


Figure 7: Breadboard implementation of Task 3.



Figure 8: DMM measurement showing 2.4836 V.

6 Discussion & Conclusion

The experiment successfully demonstrated the loading effect. In Task 1, the high output impedance of the divider caused the measurement to be inaccurate. In Task 2, the BJT buffer acted as an impedance transformer but introduced a V_{BE} drop. Finally, Task 3 demonstrated that a complementary pair can achieve unity gain while effectively cancelling the DC offset.